

3.1 Technology in Handsets



- 1 RF TX and RX amplifiers:** These handle the signals in and out of the antenna.
- 2 The inner circuit:** A lot of chips are placed on the PCB. Each has an assigned function.
- 3 RF and Power:** It handles the power management and recharging of the cell phone; it also deals with the reception of FM channels in phones with that feature.
- 4 Microprocessor and memory:** Both these chips co-ordinate the various functions of the cell phone and are responsible for communicating with the base station.
- 5 DSP:** The Digital Signal Processor is a customised processor that performs signal manipulation computations at high speeds. It also takes care of signal compression and decompression.
- 6 D/A and A/C chips:** These chips work in conjunction with each other and are responsible for the conversion of analogue signals to digital and vice versa.

The above image depicts the internals of a basic mobile phone:

Display Types

Not only do mobile phone displays show caller information, menu options, contacts, etc., mobile phone displays these days need to be capable of displaying thousands of colours so that high-end features such as images and video playback offer the desired user